



Optimization of briquetting parameters and their effects on thermochemical fuel properties of biowaste briquettes

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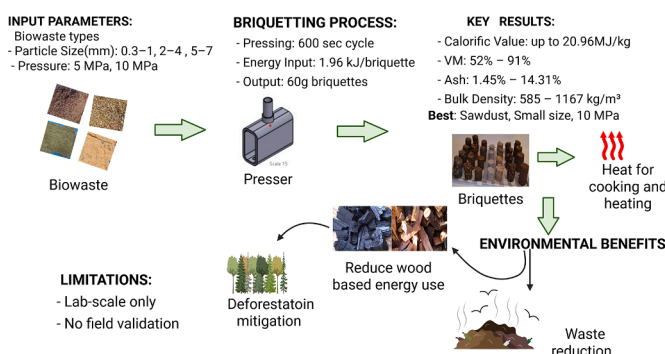
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HIGHLIGHTS

- Briquettes were produced at laboratory scale using varied particle sizes and compaction pressures.
- Biowaste type and particle size had a significant effect on fuel properties than compaction pressure.
- Briquetting reduced waste volume by 54.8%, enhancing compactness and handling.
- Combustion performance, emissions, and user-level feasibility have not yet been evaluated.

GRAPHICAL ABSTRACT



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ABSTRACT

Ethiopia faces growing environmental and energy problems because of its expanding cities, increasing dependence on biomass fuels. This study evaluated briquettes made from sawdust, coffee husk, fruit waste, flower waste and vegetable residues through different particle sizes (0.3–1 mm, 2–4 mm, 5–7 mm) and compaction pressures (5 MPa, 10 MPa) using pulped paper as a binder. The evaluation of fuel properties included density measurements, moisture content, ash content, fixed carbon, volatile matter and calorific value assessments. The combination of small sawdust particles under 10 MPa pressure produced briquettes with the highest calorific value of 20.96 MJ/kg and strong mechanical strength. The combination of small coffee husk particles under 5 MPa pressure produced briquettes with 78.05 % volatile matter and 17.62 MJ/kg calorific value. The addition of binder materials improved combustion properties and decreased the amount of volume expansion. The waste-volume reduction through briquetting reached 54.8 % which demonstrates its effectiveness as a waste reduction technique. However, this study did not include combustion emissions analysis, life cycle assessment (LCA), techno-economic evaluation (TEA), or pilot-scale field validation. Hence, the reported substitution and deforestation mitigation figures were theoretical and based solely on laboratory-scale calorific equivalence. As such,

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broader claims regarding environmental and energy system impacts should be considered preliminary and require further verification through real-world deployment and systems-level analysis.

1. Introduction

The global need for energy is increasing due to the growing population and the rise in business and industrial operations around the world. Fossil fuels such as coal, natural gas and petroleum continue to satisfy 80 % of the world's energy requirements (Sansaniwal et al., 2017), but their widespread use has caused harmful impacts on the environment. In Sub Saharan Africa, wood burning serves as the source for over 80 percent of energy needs. In Ethiopia, biomass energy supplies around 20 MWh per household annually making up 85 percent of the country's overall energy consumption (Tucho and Nonhebel, 2017; Yalew, 2022). Ethiopia's current reliance on biomass energy is causing health problems, deforestation, loss of biodiversity as well as greenhouse gas emissions and indoor air pollution (Tucho et al., 2014). These challenges highlight the need for affordable and sustainable energy solutions that are friendly to the environment. The conversion of biowaste resources into energy stands out as a solution to these issues that have gained global interest in exploring cleaner and more sustainable options as highlighted by Seboka et al. (2023).

Biowaste resources consisting of organic industrial byproducts together with municipal solid waste and agricultural residues present untapped possibilities to generate renewable energy. The annual solid waste production in Ethiopia exceeds 6 million tons and projections show this figure will reach 10 million tons by 2030 and 18 million tons by 2050 according to (Hirpe et al., 2021). The large quantity of biowaste presents favorable conditions for waste reduction and energy recovery. The poor waste management systems of Ethiopia along with other developing nations create recycling and reuse challenges for their biowastes (Ferronato and Torretta, 2019). Waste management stands as a major public issue because there are usually inadequate waste recycling and reuse practices in place (Ferronato and Torretta, 2019). The growing population, together with inadequate waste management approaches, increases environmental and infrastructure damage because they make waste collection and recycling and disposal processes more difficult (Batista et al., 2021). The practice of recovering energy from waste has emerged as a practical solution to manage waste while providing sustainable power generation.

The conversion of organic waste into bioenergy has proven effective in solving both fossil fuel scarcity and improper waste disposal issues (Guman and Wegner-Kozlova, 2020). The process of obtaining energy from waste streams is gaining popularity because it provides simple access, environmental benefits, economic savings and supports the development of a circular waste-based bioeconomy (Cong et al., 2023). The generation of renewable energy in Ethiopia and other developing countries has production challenges related to efficient and economic production due to limited financial resources and technological capabilities. The process of biomass densification through bio-briquette production establishes an efficient and cost-effective method to convert biowaste into energy. This approach helps to protect the environment while advancing sustainability by resolving waste disposal issues and providing alternative energy supply. The environmental benefits of bio-briquettes extend to multiple energy applications including heating and cooking as well as power generation. Research shows that briquettes outperform raw biomass by providing better combustion efficiency together with reduced hazardous emissions (Odugbesan and Rjoub, 2020). The process enhances waste compactness and prolongs burning duration while increasing calorific value. It also enhances waste management during treatment and storage and makes it easier to transport which improves its availability for energy generation (Tumuluru et al., 2011). Energy substitutes show promise as suitable alternatives for low-income urban and rural areas in developing nations,

because these regions have limited access to cleaner energy sources (Ferronato et al., 2023). The rising interest in converting waste to energy reflects a growing movement toward a circular waste-based bioeconomy that treats waste as a valuable resource for sustainable development (Cong et al., 2023).

The production efficiency of briquettes depends on three main factors which include biomass type and pressing pressure and biowaste particle size. Studies have demonstrated that biomass combustion properties improve when using briquettes as compressed materials increase calorific value and boost combustion efficiency (Kpalo et al., 2020; Feleke et al. 2020; Kebede, Berhe, and Zergaw 2022; Mekonen et al. 2024; Mengesha et al. 2022; Tesfaye, Workie, and Kumar 2022). They also have paid attention to the briquetting potential of woody and agricultural residues in Ethiopia but the relationship between biomass particle size and pressing pressure on fuel quality is still understudied in Ethiopia's context. The studies related to the ecological role of briquetting through waste reduction and deforestation mitigation are still limited, except for the study by Gutema et al. (2024), that analyzed floral waste. The current study examined how several types of biowaste materials such as wood-based residues, agricultural residues and municipal solid waste react to changes in pressing pressure and particle size in briquette performance. The research objectives of this study included three main goals: (i) analyzing how particle size, pressing pressure and biowaste types affect briquette physical and fuel properties, (ii) identifying the best combination of these factors to create high-quality briquettes with suitable fuel properties, and (iii) evaluating the ecological effects of briquette production on waste volume reduction and deforestation mitigation.

2. Bibliometric mapping and research gap analysis

A specific literature mapping of biowaste briquetting research in Sub-Saharan Africa was performed to define the research gap. The thematic clustering of relevant keywords and recent regional studies shows that there are a lot of work on feedstock sources and fuel property evaluation but integrated studies addressing mechanical optimization—particularly particle size and pressure interactions—and their ecological implications are rare (Table 1). This bibliometric insight supports the relevance of the present study, which combines parametric analysis with environmental considerations and provides a contribution that has not been adequately addressed in the existing regional literature.

3. Materials and methods

3.1. Biowastes collection and preprocessing

The feedstocks for briquette production were obtained from hotels and university campuses located in southern Ethiopia. Prior to biowaste collection, a detailed survey was conducted on the quantity and composition of biowaste resource generation for briquette production (Seboka et al., 2025). The availability of resources for briquette production was confirmed through this assessment. The organic waste materials used in this study include fruit waste (banana papaya and avocado), vegetable waste containing potato cabbage and onion peels and sawdust from multiple tree species including *Cupressus lusitanica*, *Pinus patula*, *Juniperus procera* and *Grevillea robusta*. Before sampling for briquetting the different parts of these biowastes were equally mixed by weight. Additionally, the flower waste was collected from Batu Sher Ethiopia Floriculture Company. The country holds the fifth position in global flower production and export while maintaining 1,700 ha of

active farms (Mebrat et al., 2022). Gutema et al. (2024), reported that five farms in Batu town produce 66,140 tons of floriculture waste throughout the year. Whereas the coffee husk samples were obtained from Nensebo Coffee Processing Cooperatives which serves as a major coffee production area in southern Ethiopia and produces internationally known Worka-Nensebo coffee. The annual estimate of coffee husk generated in Ethiopia is approximately 192,000 tons (Tolessa et al., 2022). The sun-drying of the samples took place from January to March 2024 when Ethiopia is in its dry season with the temperature ranging from 13.8 °C to 28.5 °C and the average temperature was 21.2 °C. The dried materials were transported to Norway for briquetting tests which were conducted at the Wood Laboratory of the Norwegian University of Life Sciences situated at 59.6638°N, 10.7620°E.

3.2. Sample preparation

The dried samples underwent grinding using a Retsch ZM 1000 Ultra Centrifugal Mill into different particle dimensions. The biomass was then sieved into three categories based on particle size: fine (0.3–1 mm), medium (2–4 mm), and coarse (5–7 mm), as illustrated in Fig. 1. The particle size of the raw material is important for the long-term durability of the briquettes produced.

The paper waste material from discarded office paper and newspapers functioned as the binding agent in this study. Paper waste underwent a seven-day soaking process in water before being pumped into a standardized binding solution. All trials utilized 60 g of the biomass and binder mixture which contained a 3:1 ratio of components. Before the start of the main briquetting work, a separate set of briquetting trials were carried out using small particle saw dust, to test three biomasses to binder ratios by mass 3:1, 4:1 and 5:1 (Appendix 1). These trials were used to assess the influence of binder on the physical properties and thermochemical properties of briquettes. The 10 MPa pressing pressure was held constant to evaluate the binder ratio effect. During the preliminary experiment, the reduced content of binder resulted in increased fixed carbon and ash content but reduced volatile matter and calorific value indicating a shift toward slower combustion and reduced ignition ease. Also, higher binder content (3:1) resulted in denser, more dimensionally stable briquettes. The study reported by Kebede et al. (2022) demonstrated that using a 3:1 ratio of sawdust to paper waste, coffee husk to paper waste, Khat waste to paper waste and dry grass to paper waste produced briquettes which had the highest calorific value. The study by Oladeji (2010) also demonstrated successful briquette formation and combustion behavior when using a 3:1 feedstock-to-binder ratio with agricultural residues. Thus, the selected ratio follows standard biomass densification practices and serves as an appropriate initial value for performance assessment. In this work, the biomass and binder were manually mixed until the components achieved uniform distribution. The experiment required 96 samples made from five organic waste types and three particle sizes and two pressing pressures. The production of paper waste briquettes occurred at both low- and high-pressure levels (5 and 10 MPa) which served as a control.

3.3. Briquetting machine design

The briquette pressing was undertaken using a ZwickRoell Z1200ES universal wood testing machine, which is used for testing the anatomical, physical and mechanical properties of wood. The briquette press was designed and produced in the metal workshop at Faculty of Science and Technology, Norwegian University of Life Sciences; has a rectangular hollow compression chamber of 200 mm height, 100 mm width and 12 mm wall thickness (Fig. 2). The pressing mechanism has a cylindrical extrusion at the top, which is 110 mm high, has an outer diameter of 44.5 mm and an inner diameter of 39.7 mm. The total height of the press is 121.5 mm, including the extrusion. Inside the chamber, circular pressing plates made of 2 mm perforated stainless steel are used, with two plates at the bottom for added strength. The design is efficient for compression, controlled ejection and durability, thus suitable for briquette production. The machine was set to apply forces of 5000 and 10,000 kN/m², which are equivalent to 5 MPa and 10 MPa pressure levels respectively for briquette press.

3.4. Briquette production

The mixture was subjected to hydraulic pressing using a pressing force machine in the wood laboratory. The material was placed in cylindrical molds with a diameter of 50 mm and compressed at two pressure levels: low pressure (5 MPa) and high pressure (10 MPa). The pressing process was carried out for 5 min, followed by a 5-minute holding period to ensure proper compaction (Ifa et al., 2020). After molding, the briquettes were air-dried at ambient temperature for 48 h, then oven-dried at 105 °C for 20 h, reducing the moisture content to below 8 %. The final briquettes were then characterized for their physicochemical properties, including density, volatile matter, ash content, calorific value, moisture content, and fixed carbon.

3.4.1. Physical properties

The measurements of briquette diameter, height and weight have been carried out immediately after briquetting and after 48 h. Accordingly, the volumetric expansion the briquettes were calculated using the equation (Ramírez-Ramírez et al., 2021).

$$VE(\%) = \left[\frac{(V_2 - V_1)}{V_1} \right] * 100 \quad (1)$$

where EV is Volumetric expansion (%), V_1 is volume of the briquette immediately after compaction (mm³) and V_2 is volume of briquette after conditioning time (mm³).

The bulk density of briquette was calculated by weighing the mass of the sample and measuring its volume and then using the following formula:

$$Bulk\ density \left(\frac{kg}{m^3} \right) = \frac{mass\ of\ sample(kg)}{volume\ of\ sample(m^3)} \quad (2)$$

Table 1
Bibliometric analysis of biomass briquetting research in Sub-Saharan Africa.

Cluster	Keywords	Thematic focus	Literature coverage	Gap identified	References
Cluster 1	Biomass, Briquetting, Calorific Value, Fuel Properties	Technical evaluation of briquette fuel performance	Moderate (~50 %)	Limited multi-parameter performance analysis (e. g., emissions, density, durability beyond calorific value)	(Kpalo et al., 2020; Tesfaye et al. 2022)
Cluster 2	Agricultural Waste, Waste Management, Sub-Saharan Africa	Feedstock sources and local context	High (≥60 %)	Insufficient optimization or comparison across diverse biomass types	(Feleke et al., 2020; Mitchual et al., 2013)
Cluster 3	Sustainability, Energy Access, Rural Development	Socioeconomic and policy impacts	Moderate (~40 %)	Weak integration with technical metrics (e.g., stove compatibility, emissions)	(Dujmović et al., 2022; Njenga et al., 2019)
Cluster 4	Pressure, Particle Size, Compaction, Ash Content	Mechanical and ecological design optimization	Low (<15 %)	Very few studies explore combined parametric effects or environmental trade-offs	(Ibitoye et al., 2022; Mengesha et al., 2022)

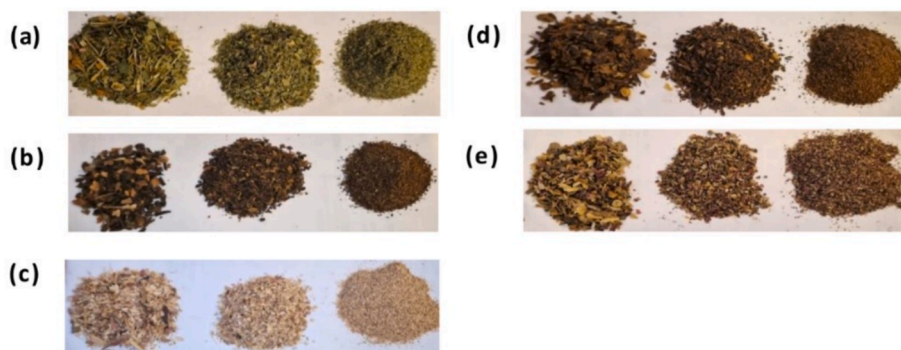


Fig. 1. Biowastes prepared for briquetting at 3 levels of particle size; (a) flower waste, (b) fruit waste, (c) sawdust, (d) coffee husk and (e) vegetable waste.

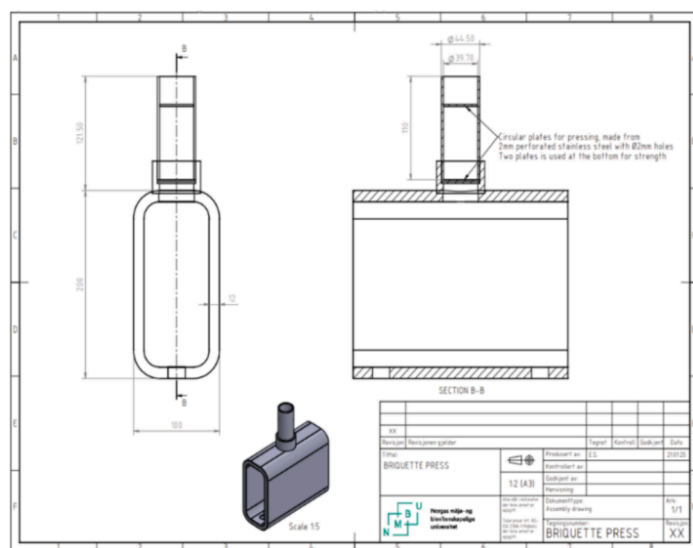


Fig. 2. The 3D design and dimensions of the briquette press.



The mechanical durability of the produced briquettes underwent standard testing procedures to achieve consistent results. The ASTM E9-09 (Reapproved 2019) standard was used to measure compressive strength during compression testing at room temperature. The ASTM D440-07 (Reapproved 2012) standard was used to evaluate impact resistance through drop shatter tests. The ISO 17831-2:2015 standard was used to determine the mechanical durability of briquettes through abrasion resistance testing. Standardized tests provided a complete evaluation of briquette integrity during storage and handling operations.

3.4.1.1. Proximate analysis. Proximate analysis of raw biowaste materials and briquettes served to evaluate essential parameters affecting briquette combustion efficiency and quality. The proximate analysis of briquette samples was performed in triplicates to achieve accurate results which produced average values for reporting. The briquette samples underwent standard ASTM procedures to determine their moisture content and volatile matter and ash content and fixed carbon. The moisture content analysis followed ASTM E1756-08 (ASTM E871-82, 2015). The volatile matter analysis followed ASTM E872-82 (ASTM, 2011) procedures which involved heating moisture-free samples in a covered crucible at 950 °C for 7 min. The ash content measurement followed ASTM E1755-01 (ASTM, 2015), by burning the material at 760 °C until a permanent residue formed. The remaining percentage after subtracting moisture content from volatile matter and ash from 100 % gave the fixed carbon content (Narzary et al., 2020). The

standardized methods delivered exact and standardized measurements of the fundamental fuel properties in biomass briquettes.

3.5. Calorific value determination

The calorific value of briquette was determined by using 6400 Bomb Calorimeter (Parr Instrument Company, Illinois, USA). One gram of the sample was palletized, placed in a sample holder (crucible), and then transferred to a steel capsule from the bomb calorimeter. It is determined by using standard procedure as per ISO 9831. The samples are completely incinerated in a closed system with an excess of oxygen (bomb calorimeter). During combustion, the chemical bonds between the atoms change and when all the material has been burned, all the organic molecules will be converted to mainly CO₂ and H₂O, cf. combustion of methane in excess oxygen. In addition, there was an inorganic residue that is not incinerated (ISO 9831, 1998).

3.6. Statistical analysis

The experiments were conducted in triplicate and the data analysis included descriptive statistics (mean, standard deviation, standard error) and inferential statistics such as multiple linear regression under General Linear Model (GLM), analysis of variance (ANOVA) and paired *t*-test to study the influence of particle size, pressing pressure and biowaste types on the physicochemical fuel properties of the briquettes. A predictive model-based optimization approach was used to determine

the optimal combination of briquette production parameters by applying the coefficients obtained regression model. The estimated regression coefficients were used to create predictive models for each fuel property. The final model did not include interaction effects because they were found statistically insignificant ($p > 0.05$). The statistical analysis was performed with the General Linear Model procedure in Minitab 22.2.2.0 software.

4. Results and discussions

The findings of this study were systematically organized into six distinct sections. Initially, the raw biomass materials were presented with detailed information on their chemical composition and proximate analysis. Following this, the briquettes produced were characterized by their chemical and physical properties. The third section focused on modeling how pressure pressing, particle size, and the types of biowaste could influence the fuel's characteristics. The fourth segment predicts the parameter resulted in high-quality briquettes suitable for fuel applications. The fifth section of this study focused on investigating the role of briquetting in waste reduction and deforestation mitigation. The final section summarized the key findings and discussed their implications for advancing sustainable fuel development.

4.1. Biochemical composition of raw biowastes

The raw biomasses chosen for briquette production present diverse chemical compositions which determine their combustion properties and binding capabilities (Table 2). The glucan content reaches its highest level in vegetable waste (VW), 31.0 % followed by paper at 28.5 %. The glucan content of flower waste (FW) stands as the lowest at 7.7 %. The hemicellulose-containing substances xylan and mannan exist abundantly in sawdust (SD) and paper which helps maintain structural integrity. The lignin concentration reaches its highest point in SD (40.5 %) while VW contains the lowest lignin content (16.0 %) which determines binding and combustion stability. The chemical composition of the raw biomasses used for briquette production reveals varying levels of nitrogen and carbon. Flower waste (FW) has the highest nitrogen content at 2.4 %, while sawdust (SD) has the lowest at 1.2 %. Vegetable waste (VW) has a carbon percentage of 37.5 %, whereas sawdust boasts a higher carbon content of 46.8 %. The carbon-to-nitrogen (C: N) ratio further illustrates the need for certain biomass types to be mixed to achieve optimal performance; FW has the lowest C:N ratio at 17.9, and SD has the highest at 40.1.

4.2. Proximate analysis of raw biowastes

The briquetting process utilized multiple types of raw biowastes which included flower waste, vegetable waste, coffee husk as well as sawdust and paper waste for binding purposes. The chosen materials were suitable because they existed readily and demonstrated fuel

potential. The selected biowastes underwent fuel property evaluation to determine their potential as biomass fuel sources before briquetting. The lowest volatile matter (VM) content exists in Fruit waste (FrW) with 63.8 % which indicates stable combustion. The high volatile matter content in sawdust (SD) and flower waste (FW) reaches 85.4 % and 78.7 % respectively which indicates fast ignition and high combustion reactivity (Table 3). The combustion performance of paper waste (PW) and vegetable waste (VW) was better due to their lowest MC values which amount to 3.7 % and 3.8 % respectively. FW possesses the highest MC value of 6.2 % which might need pre-drying to achieve maximum energy output. Stolarski et al. (2013), reported that effective briquette production requires raw material moisture between 6 % and 23 %. The formation of rough briquettes occurs when moisture content is low but high moisture requires additional energy expenses for drying.

The maximum fixed carbon (FC) percentage required for sustaining combustion exists in PW (13.0 %) and FrW (12.3 %) thus indicating better char combustion performance, but SD (7.1 %) and FW (7.8 %) show lower FC values and rely on volatile combustion. In terms of energy value, SD (19.2 MJ/kg) and FrW (18.4 MJ/kg) demonstrate the highest calorific value (CV) among the samples which indicates their strong potential as biofuels. The calorific value of the remaining samples extends between 15.0 MJ/kg and 17.5 MJ/kg. The analysis of volatile matter and fixed carbon reported by Gutema et al. (2024), matched the current study since they reported 72.3 % volatile matter and 10 % fixed carbon from their study on raw biomass from flower waste. Their study achieved a calorific value of 18.3 MJ/kg which was higher than the present investigation. Direct combustion faces a major efficiency problem in heating performance. According to Li et al. (2017) The problems with biomass material handling and storage become more severe because of their high moisture content and low bulk density. The conversion of biomass into briquettes provides an effective solution to overcome these problems (Kaliyan and Vance Morey, 2009).

4.3. Theoretical basis and quantitative modelling of particle size and pressure effect

The effects of particle size and compaction pressure on thermal behavior and densification quality can be understood through several interconnected engineering mechanisms. The high surface-area-to-volume ratio of smaller particles enables better heat and mass transfer efficiency and reduces internal thermal gradients and enhances thermal decomposition uniformity (Chen et al., 2020; Guo et al., 2020) The increase of compaction pressure decreases porosity while changing pore connectivity which limits convective flow and volatile transport according to capillary transport theory (Kaliyan and Vance Morey, 2009). The densification process under elevated pressure strengthens inter-particle bonds through van der Waals forces and solid bridge formation and mechanical interlocking mechanisms which become more effective because of increased contact area and plastic deformation (Anukam

Table 2

Overview of the chemical composition of raw biomasses used in briquette production.

Chemical compositions (%)	CH	FW	Sd	VW	FrW
Glucan	11.2	7.7	23.3	31.0	11.8
Arabinan	2.1	0.9	0.4	1.1	0.8
Galactan	1.4	1.1	1.7	2.4	0.9
Xylan	0.8	0.9	3.4	1.1	0.9
Mannan	1.0	0.5	3.6	0.8	0.9
Lignin	32.3	31.7	40.5	16.0	33.9
Nitrogen	1.7	2.4	1.2	1.8	1.8
Carbon	41.2	42.9	46.8	37.5	42.5
Sulfur	0.1	0.1	0.1	0.2	0.2
Ammonium-N*	0.2	0.4	0.2	1.1	0.3
C:N ratio*	24.6	17.9	40.1	21.2	29.3

* g/kg.

Table 3

Summarizes the mean values and standard errors (SE) for several important fuel properties—across raw biomass samples.

Biowaste sample	VM%	Ash %	MC%	FC %	Calorific value (MJ/kg)
CH	73.9 ± 0.5	10.9 ± 0.4	5.6 ± 1.4	9.5 ± 2.1	17.5 ± 0.1
FrW	63.8 ± 0.5	12.9 ± 1.1	5.1 ± 0.4	12.3 ± 1.1	18.4 ± 0.0
FW	78.7 ± 2.2	7.3 ± 0.3	6.2 ± 1.9	7.8 ± 1.5	17.2 ± 0.1
PW	73.8 ± 1.8	9.6 ± 0.2	3.7 ± 0.1	13.0 ± 2.0	15.0 ± 0.0
SD	85.4 ± 1.5	2.0 ± 0.1	5.5 ± 0.9	7.1 ± 0.7	19.2 ± 0.0
VW	75.1 ± 1.8	9.3 ± 1.1	±0.5	11.8 ± 0.7	± 0.1

et al., 2021; Karamchandani et al., 2019). In this paper, the quantitative modeling framework was also established to support a mechanistic interpretation of briquette fuel behavior, capturing the coupled effects of particle size and compaction pressure on heat and mass transfer, pore structure development, and devolatilization kinetics.

4.4. Devolatilization kinetics

The combustion process of biomass devolatilization follows first-order Arrhenius kinetics which express the reaction rate constant $k = A \cdot e^{-E_a/RT}$ where k is the reaction rate constant (s^{-1}), A is the pre-exponential factor, E_a is the activation energy (kJ/mol), R is the universal gas constant (8.314 J/mol·K), and T being the absolute temperature in Kelvin. The activation energy (E_a) for lignocellulosic biomass devolatilization ranges between 45 kJ/mol and 150 kJ/mol according to Anukam et al. (2021) and Guo et al. (2020) depending on feedstock composition and particle size. The devolatilization kinetics of sawdust at 650 K pyrolysis temperature is demonstrated by calculating the rate constant k using different activation energy values. For $E_a = 60$ kJ/mol, $k = \frac{10^{6.6} \cdot e^{-60000}}{8.314 \cdot 650} = 4.91 s^{-1}$, For $E_a = 90$ kJ/mol, $k = 0.20 s^{-1}$. The devolatilization rate becomes faster when E_a values are lower as seen in small particle sawdust briquettes which produce 90.87 % VM and fast ignition and high calorific value. The combination of empirical trends and kinetic modeling shows that smaller particle sizes decrease effective E_a through enhanced reactive surface area and heat transfer efficiency which results in faster volatile release and improved combustion efficiency.

4.5. Capillary transport and pore network analysis

During the densification process, biomass undergoes compaction, leading to a reduction in pore size and alterations to its internal flow pathways. These structural changes affect the movement of moisture and volatiles within the briquette, particularly under drying and combustion conditions. The Lucas–Washburn equation $L = \left(\frac{r \cos \theta}{2\mu} \right)^{1/2} \cdot \sqrt{rt}$ describes capillary-driven moisture transport, where L is the penetration depth, r is the pore radius and γ is the liquid surface tension (0.072 N/m for water) and μ is the dynamic viscosity (~ 0.001 Pa·s), θ is the contact angle (typically 30° for biomass) and t is time. This study used the findings of (Kaliyan and Vance Morey, 2009), to determine effective pore radius through their analysis of compaction levels and particle sizes. The pore radius of briquettes produced at 5 MPa with 2–4 mm particle size ranges between 15 and 20 μm because these briquettes have a less dense structure with bigger voids. Highly compacted briquettes (10 MPa) containing small particles (0.3–1 mm) have pores that measure between 4 and 6 μm because their high packing density creates small voids. The calculated moisture penetration distance shows a 50 % reduction in the more compacted briquettes with smaller particle sizes. The experimental results support this prediction because sawdust briquettes made at higher pressure with finer particles showed a lower moisture content of 1.69 % compared to 3.26 % for briquettes made at lower pressure with coarser particles. The reduced pore radius creates barriers to capillary flow while increasing the path length which restricts moisture transport and enhances both drying and combustion efficiency.

4.6. Thermal diffusion and heat transfer analysis

The Fourier number $Fo = \alpha t/L^2$ was used to evaluate how particle size affects heat transfer where α represents thermal diffusivity and t represents heating time and L represents the characteristic length (particle radius). The Fourier number for small particles with a radius of 0.25 mm reaches 1.6 when thermal diffusivity equals $1.0 \times 10^{-7} m^2/s$ and heating last for 10 s. The Fourier number of 0.11 for large particles with a radius of 3 mm shows that these particles experience slower

internal heat diffusion and thermal gradients. The observed improvement in volatile matter release and calorific value in small-particle briquettes (e.g., 90.87 % VM and 20.96 MJ/kg for sawdust) can be explained by the rapid internal heating which promotes devolatilization. The findings match earlier research which demonstrated that smaller biomass particles enhance thermal transfer and pyrolysis efficiency (Guo et al., 2020; Kaliyan and Vance Morey, 2009).

4.7. Pore structure evolution and interparticle bonding

The quantitative analysis of bulk density and volumetric expansion (Table 4) showed that briquettes compressed at 10 MPa had significantly reduced expansion and higher post-conditioning densities. For instance, coffee husk briquettes showed a density drop from 1167.3 kg/m³ to 606.98 kg/m³ with ~ 24.5 % expansion. Smaller particle sizes enable tighter packing and improved van der Waals forces, solid bridge formation, and mechanical interlocking—contributing to stronger bonding. The reduced void ratio also improves combustion consistency.

4.8. Physical properties

4.8.1. Bulk density and volumetric expansion

The different biowaste briquettes showed considerable variation in bulk density, density after 48 h, and volumetric expansion. The coffee husk (CH) briquettes had the highest initial bulk density (1167.3 ± 81.1 kg/m³), followed by the vegetable waste (VW) briquettes (1138.8 ± 146.5 kg/m³) and the fruit waste (FrW) briquettes (1072.8 ± 68.8 kg/m³). On the other hand, the sawdust (SD) briquettes had the lowest bulk density (585.0 ± 74.3 kg/m³) (Table 4). The result of this finding was higher than the briquette bulk density of 970 kg/m³ reported by Tesfaye et al. (2022) for briquettes made from coffee husk. All the briquette types showed a decrease in density after 48 h, with CH briquettes having the highest density (606.98 ± 16.68 kg/m³) and SD briquettes the largest decrease (390.44 ± 62.38 kg/m³). The FW and FrW briquettes had the highest volumetric expansion (27.58 ± 11.97 %), which indicate they had the highest structural instability and moisture absorption, while the SD briquettes had the lowest expansion (11.86 ± 9.50 %), that implicate the best dimensional stability.

The paired t -test showed that a significant decrease in density occurred throughout time because it revealed a statistically significant difference between the initial bulk density and the density after 48 h ($t = 12.11$, $p = 0.001$). The results show that all types of biowaste materials experience structural instability or compaction loss which would impact their long-term performance and durability.

4.8.2. Effect of particle size, pressure and biowaste types on the initial density

The two-way ANOVA results indicated that biowaste type and pressing pressure significantly affect briquette density after 48 h, but

Table 4

Physical properties of briquettes produced from different organic wastes (values were presented as mean $\pm$ SD; N = sample size).

Biowaste samples	Bulk density (kg/m ³)	Density after 48 h (kg/m ³)	Volumetric expansion (%)
CH	1167.3 $\pm$ 81.1 (N = 4)	606.98 $\pm$ 16.68 (N = 4)	24.47 $\pm$ 2.05 (N = 4)
FrW	1072.8 $\pm$ 68.8 (N = 6)	791.26 $\pm$ 59.15 (N = 6)	27.31 $\pm$ 5.56 (N = 6)
FW	823.8 $\pm$ 154.2 (N = 6)	540.88 $\pm$ 134.26 (N = 6)	27.58 $\pm$ 11.97 (N = 6)
PW	830.6 $\pm$ 155.3 (N = 2)	593.58 $\pm$ 17.08 (N = 2)	20.11 $\pm$ 10.93 (N = 2)
SD	585.0 $\pm$ 74.3 (N = 6)	390.44 $\pm$ 62.38 (N = 6)	11.86 $\pm$ 9.50 (N = 6)
VW	1138.8 $\pm$ 146.5 (N = 6)	796.63 $\pm$ 45.96 (N = 6)	24.96 $\pm$ 12.92 (N = 6)

particle size does not. The density of FW decreased while FrW increased density significantly and CH showed the greatest density increase among the biowaste types ($p = 0.002$ and $p = 0.000$ respectively). The briquette density showed a significant decrease in relation to increasing pressing pressure (Fig. 3c). The F-value and t -values showed no significant effect ($p > 0.05$) of particle size on density.

4.8.3. Effect of particle size, pressure and biomass type on volume expansion

Briquettes demonstrated different volumetric expansion patterns based on their composition and compaction pressure and particle sizes. The dimensional stability and compaction behavior of briquettes resulted in the lowest expansion rates for both sawdust (SD) and paper waste

(PW) at 5 MPa and 10 MPa (Fig. 3d). The structural integrity of vegetable waste (VW) deteriorated when subjected to elevated pressures which led to increased expansion. The results showed that particle size affected expansion behavior differently among biowaste types because SD kept its expansion rates minimal across all particle sizes, yet FW and VW showed increased expansion mainly with medium and large particles (Fig. 3e). in line with this, Mitchual et al. (2013), reported that the relaxed density of briquettes made from tropical hardwood sawdust depends on species type and particle size and compacting pressure.

4.8.4. Mechanical durability assessment

The mechanical and durability properties of briquettes showed significant differences based on the type of biowaste and its particle size, as

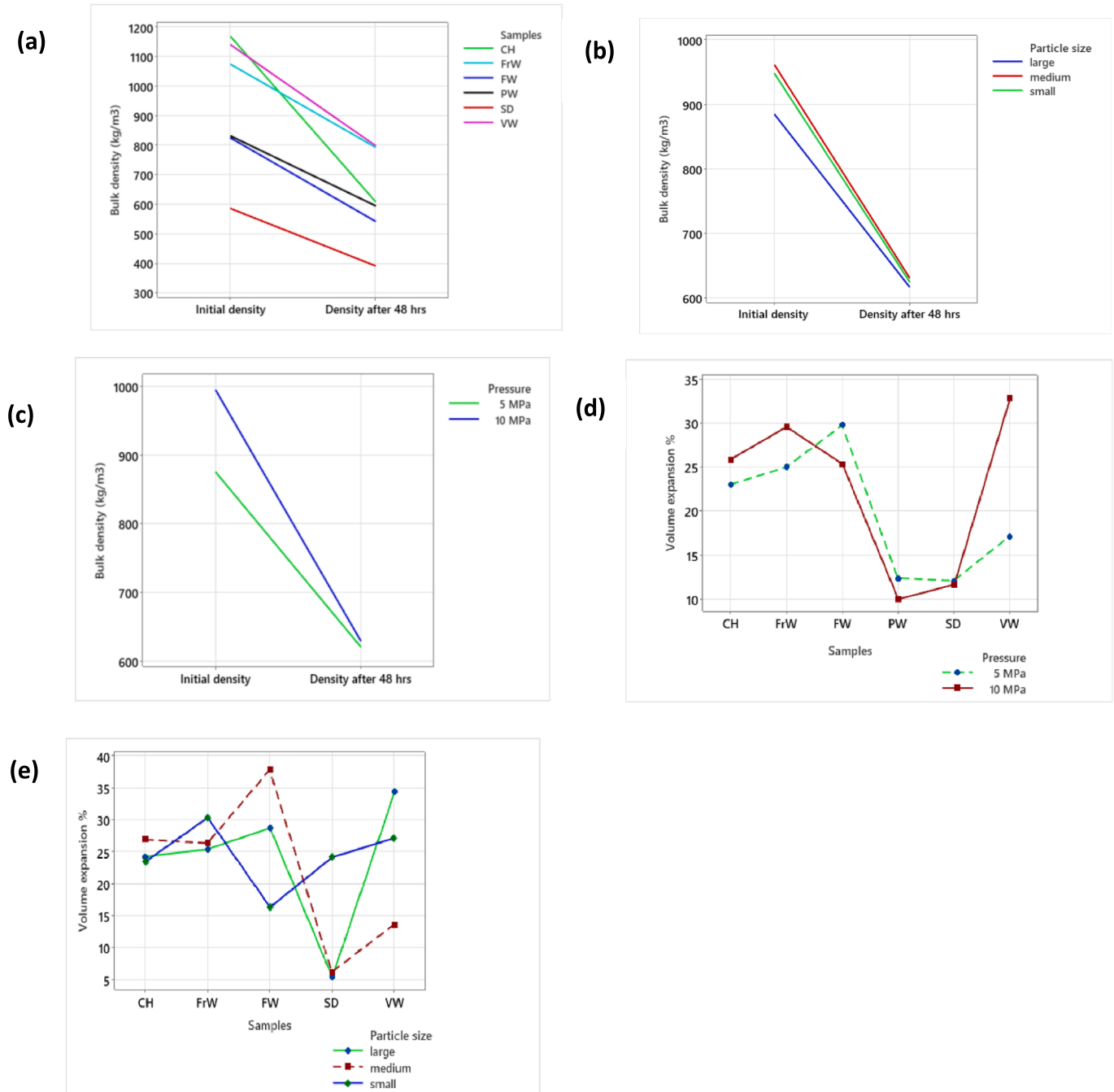


Fig. 3. Bulk density changes across biowaste samples, particle size and pressure (as shown in figure a, b and c respectively); the volumetric expansion behaviour of samples under different briquette parameters: (d) pressing pressure, (e) particle sizes.

well as the applied pressure and moisture content. The compressive strength of sawdust briquettes was 3.37 MPa, while their abrasion loss remained below 10 %, and it survived 4–5 drops due to their small-particle size and suitable moisture level of 7.97 % (Appendix 3). The strength of coffee husk and flower waste briquettes ranged between 1.50 and 2.90 MPa, but its drop resistance was reduced by using lower pressure or insufficient moisture. The performance of fruit and vegetable waste briquettes was poor because their strengths were less than 1.2 MPa, and they showed high abrasion loss and broke after minimal-impact. The results demonstrated that briquette quality depends on using fine particles together with proper compaction pressure and suitable moisture levels.

4.9. Ultimate analysis of briquette and binder

The elemental analysis of briquettes and pulped paper binder shows that carbon (C) content was highest in sawdust briquettes (48.1 %) and lowest in vegetable waste (38.3 %) which is due to the variation in lignocellulosic composition among the biomass types (Appendix 2). The hydrogen (H) content was narrow, ranging from 6.1 % to 6.6 % which indicates that the energy releasing potential is consistent across the samples. The nitrogen (N) levels were generally low ranging from 0.1 % in the binder to 2.5 % in flower waste which is favorable for minimizing NO_x emissions during combustion. The sulfur (S) content was either very low (0.1–0.2 %) or not detected in sawdust which suggests that there is minimal environmental impact from sulfur emissions. The pulped paper binder had a comparable carbon (43.1 %) and hydrogen (6.1 %) content to the biomass types which confirms its thermochemical compatibility and active role in combustion. Its low nitrogen and sulfur levels further support its contribution to cleaner fuel performance.

4.10. Proximate analysis of briquettes

The fuel properties of briquettes produced from five different bio-waste sources with their particle size categories were presented in Table 5. The volatile matter (VM) percentage ranged from 52.39 % (vegetable waste with large particles) to 90.87 % (sawdust with small particles). Particles with smaller sizes were typically found to have higher VM content, which is beneficial for combustion efficiency. This agrees with Chukwunke et al. (2020), who reported that sawdust from Gmelina and Oak tree species had high VM values of 83.1 % and 81.6 %, respectively. The ash content of the feedstocks caused the ash concentration to differ significantly among the biomass types, with the highest content observed in large particle vegetable waste (14.31 %) and the lowest in small particle flower waste (2.85 %). Brunerová et al. (2017) also observed low ash percentage in briquettes produced from tropical fruit waste (3.67 %–5.05 %), which in agrees with the current study that found lower ash content improves combustion quality. The moisture content (MC) of smaller particle sizes was lower, with sawdust having the lowest MC of 1.69 %. This could increase combustion efficiency. This was in line with Ramírez-Ramírez et al. (2021), who found that sawdust briquettes in Mexico had a moisture content ranging from 4.1 % to 7.2 %. The paper waste had a calorific value of 15.33 MJ/kg, a VM of 76.45 %, and an ash concentration of 9.51 %. Yusuf Kpalo and Faiz Zainuddin (2020), in their study found that briquetting is a sustainable alternative to fossil fuels since densified biomass can improve combustion characteristics by 20 % and significantly reduce the emission of greenhouse gases, nitrogen oxides (NO_x) and sulfur dioxide (SO₂).

Different biomass types exhibited varying patterns in fixed carbon (FC) and calorific value (CV). Fruit waste with small particles had the lowest FC level at 5.36 %, whereas vegetable waste with large particles had the highest FC level at 21.06 %. Fruit waste and sawdust demonstrated significant energy potential, with calorific values ranging from 13.03 MJ/kg (large particle flower waste) to 20.96 MJ/kg (small particle sawdust). These findings are consistent with previous research. For example, briquettes made from tropical fruit waste had calorific values

Table 5

The mean and standard deviation (SD) of briquette properties.

Biomass types	Particle size	VM% mean ± SD	Ash (%) mean ± SD	MC (%) mean ± SD	FC (%) mean ± SD	Calorific value (MJ/kg) Mean ± SD
Sawdust (SD)	Large	69.82 ± 1.33	8.21 ± 1.34	7.52 ± 3.03	14.44 ± 0.36	15.22 ± 0.28
	Medium	88.03 ± 2.22	2.02 ± 0.10	3.26 ± 0.34	6.69 ± 1.78	19.25 ± 0.78
	Small	90.87 ± 0.62	2.52 ± 0.62	1.69 ± 0.45	4.92 ± 0.44	20.96 ± 0.87
Flower waste (FW)	Large	58.76 ± 4.30	12.22 ± 0.95	11.07 ± 0.86	17.95 ± 2.49	13.03 ± 0.57
	Medium	74.82 ± 4.94	6.59 ± 0.16	4.91 ± 0.30	13.68 ± 4.49	16.07 ± 0.08
	Small	90.54 ± 3.31	2.85 ± 0.42	2.45 ± 0.79	4.16 ± 4.51	17.93 ± 0.77
Coffee husk (CH)	Medium	66.16 ± 4.43	11.76 ± 1.56	10.36 ± 0.45	11.73 ± 5.54	16.37 ± 0.23
	Small	80.12 ± 5.58	6.21 ± 3.09	5.91 ± 1.06	7.75 ± 1.43	18.50 ± 2.09
Fruit waste (FrW)	Large	68.16 ± 1.66	11.01 ± 1.23	8.38 ± 3.00	12.45 ± 2.58	14.98 ± 0.25
	Medium	88.06 ± 1.49	4.31 ± 0.79	2.28 ± 1.37	5.36 ± 0.67	19.87 ± 1.08
	Small	64.01 ± 1.62	12.12 ± 0.47	5.94 ± 0.93	17.93 ± 2.09	15.90 ± 0.14
Vegetable waste (VW)	Large	52.39 ± 2.71	14.31 ± 3.26	12.24 ± 2.95	21.06 ± 2.39	15.19 ± 1.40
	Medium	69.54 ± 2.18	9.53 ± 0.49	7.65 ± 0.62	13.28 ± 3.29	17.89 ± 0.27
	Small	59.70 ± 6.02	10.43 ± 2.67	9.71 ± 2.16	20.15 ± 1.19	14.40 ± 0.76
Paper (PW)	Small	76.45 ± 3.36	9.51 ± 0.23	3.93 ± 0.06	10.11 ± 3.53	0.00

of 17.6–18.22 MJ/kg, according to Brunerová et al. (2017), while the finding of Arulprakasajothi et al. (2020), reported that fruit and vegetable briquettes had calorific values ranging from 15.28 to 17.85 MJ/kg. Similarly, the present study found that sawdust briquettes had calorific values of 20.5–22.8 MJ/kg, volatile matter (VM) of 82.9 %–90.7 %, and fixed carbon of 8.9 %–16.4 %. According to Arulprakasajothi et al., (2020), the density of sawdust briquettes ranged from 813.9 to 1014.4 kg/m³. Additionally, Narzary et al., (2020), discovered that briquettes made from dry leaves, grass, and agricultural wastes including rice husk and plant stalks have the same fuel qualities as those identified in this study, supporting the effectiveness of biomass briquettes.

The pulped paper binder showed active thermal properties in briquettes made from different biowastes throughout this research. The binder contains 43.1 % carbon and 6.1 % hydrogen which produces a calorific value of 15.0 MJ/kg that matches or exceeds the values of several raw biomass materials including large-particle flower waste (13.03 MJ/kg) and vegetable waste (15.19 MJ/kg). The high volatile matter content of 73.8 % indicates that the binder plays a significant role in devolatilization processes. The binder helps release volatile substances early during combustion which leads to improved ignition and flame spread especially when used with low-volatile biomass materials such as large-particle vegetable waste (52.39 % VM) or flower waste (58.76 % VM). The binder maintains a suitable fixed carbon content of 13.0 % and ash content of 9.6 % which supports prolonged combustion without creating excessive ash accumulation. The binder functions as a matrix component in briquettes to improve combustion stability and fuel uniformity through its dual role of releasing volatiles and

strengthening briquette structure. The binder actively affects thermal behavior and ash dynamics and fuel performance which supports its thermochemical evaluation and confirms its importance in enhancing briquette quality.

4.11. General linear model analysis of briquette fuel properties model diagnostics and residual analysis

The diagnostic plots of volatile matter (VM%), Calorific value (CV), Fixed carbon (FC) models were presented in Fig. 4. The normal probability plot of standardized residuals (Fig. 4a, c and e) showed that the residuals are in good agreement with the reference line, thus the normality assumption is fairly met. The residuals vs. fitted values plot (Fig. 4b, d and f) shows no obvious pattern, thus the assumption of

homoscedasticity is met. The residuals are randomly scattered around zero with no systematic trends, thus the model does not suffer from heteroscedasticity or non-linearity. These diagnostics, therefore, confirm the appropriateness of the model for predicting VM%, CV MJ/kg and FC%.

4.12. Volatile matter

The model analysis indicated that the volatile matter (VM%) content was unaffected by the applied pressing pressure ($p = 0.492$) but both sample type ($p = 0.005$) and particle size ($p = 0.001$) produced statistically significant results. The model fit was moderate because it explained 66.80 % of the total variation ($R^2 = 0.668$). The VM% of sawdust (SD) exceeded that of other biowastes in every sample type (p

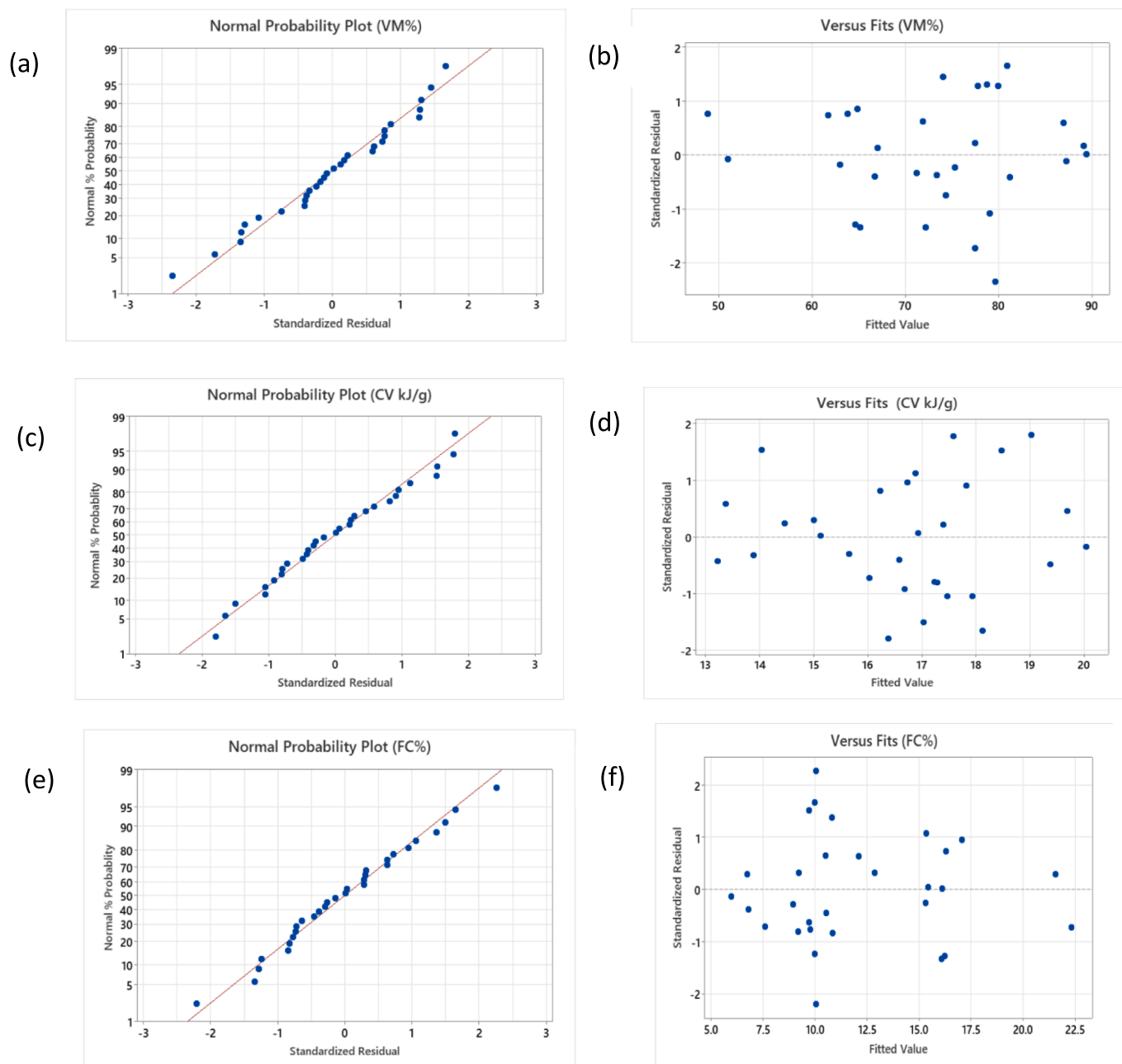


Fig. 4. The standardized residuals of (a) Normal probability plot of residuals of volatile matter (VM%) and (b) Residuals vs. Fitted values plot for VM% model; (c) Normal probability plot of residuals of calorific value (CV), (d) Residuals vs. Fitted values plot for CV model; (e) Normal probability plot of residuals of fixed carbon (FC %), (f) Residuals vs. Fitted values plot for FC % model.

= 0.003). The medium-sized particles produced higher VM% values ($p = 0.018$) than small particles but large particles generated significantly lower VM% values ($p < 0.001$). This finding was matching previous research results but contain small discrepancies. The study conducted by Wilson (2016), demonstrated that maize cob briquettes did not exhibit any changes in fuel properties when biomass particle size and pressing pressure were analysed. Song et al., (2010) observed that barley and canola and oat and wheat straw briquettes experienced significant changes with pressing pressure and temperature, but the particle size made no difference. Research discrepancies may stem from differences between biomass materials together with varying compression parameters and binder materials in the studies.

Different biowaste samples with large medium and small particle sizes exhibit VM% variability which is shown in Fig. 5 (g). Small particles tend to produce better breakdown and combustion efficiency because of their high surface area-to-volume ratio increases VM%. The graph in Fig. 5 (h) shows VM% variations between the two pressure values of 5 MPa and 10 MPa. VM% tends to increase with pressure application but the degree of this relationship depends on the type of biowaste. The VM% of VW remains stable, yet SD and PW demonstrate significant increases.

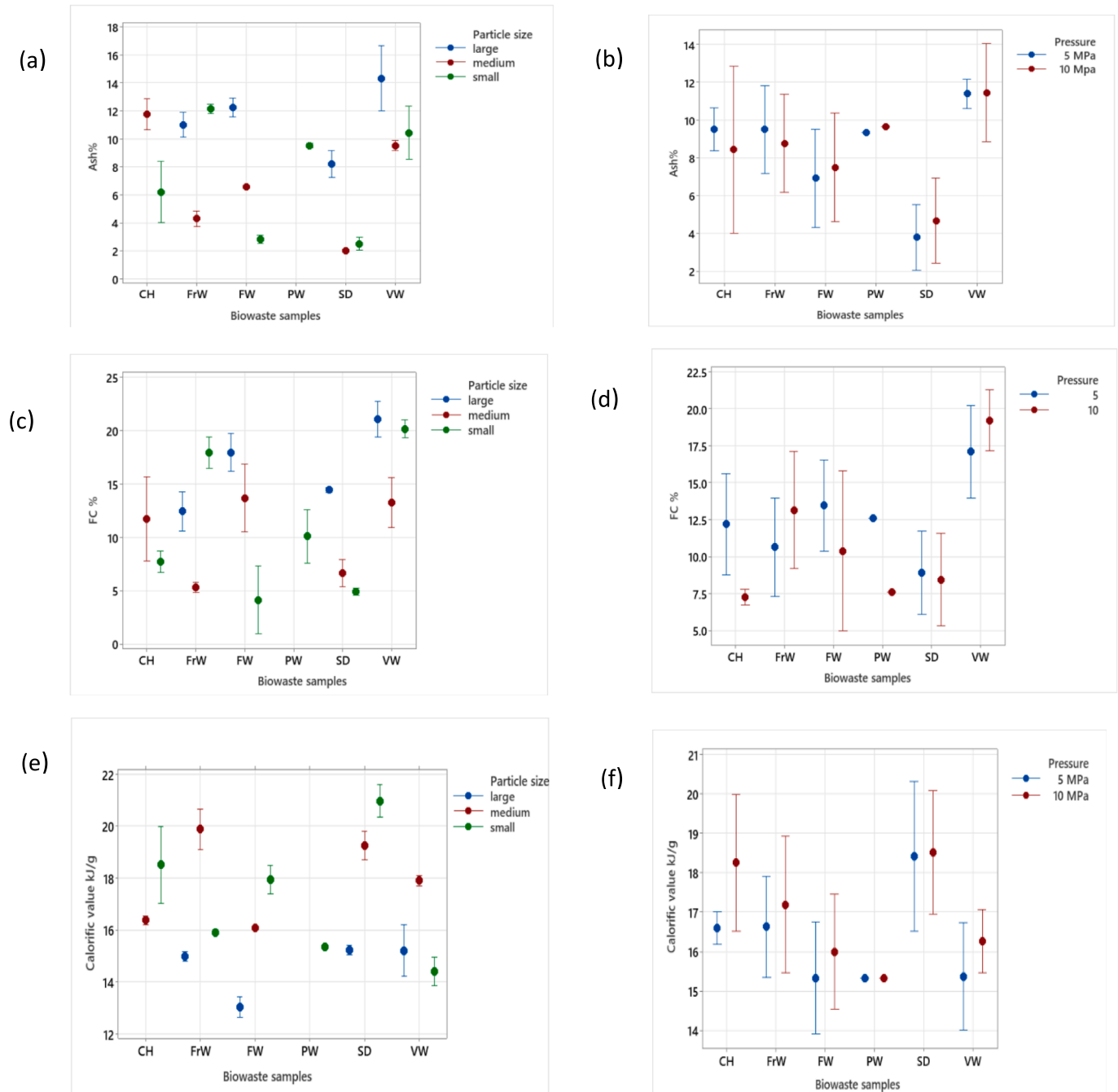


Fig. 5. Interval plots showing the influence of particle size and pressing pressure on three key fuel properties of biowaste samples: (a, b) Ash content (%), (c, d) Fixed carbon content (%), (e, f) Calorific value (MJ/kg), and (g, h) Volatile Matter (%). Subfigures (a), (c), (e) and (g) show variations across particle size (large, medium, and small), while (b), (d), (f) and (h) illustrate effects of pressing pressure (5 MPa vs. 10 MPa).

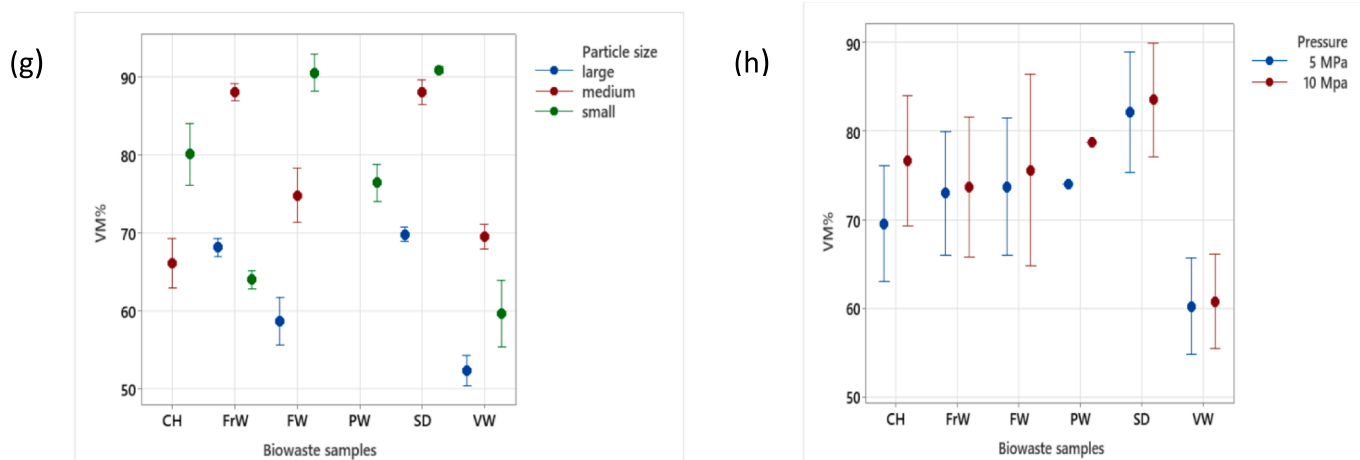


Fig. 5. (continued).

4.13. Moisture content

The biowaste samples showed coffee husk (CH) contained the most moisture ($p = 0.003$) while sawdust (SD) contained the least ($p = 0.002$) thus demonstrating how sample composition affects moisture retention. The results showed that bigger particles maintained higher moisture levels ($p = 0.000$) and medium-sized particles contained less moisture than smaller particles ($p = 0.013$). According to the GLM output data, moisture content was significantly impacted by the types of biowaste sample ($p = 0.000$) and particle size ($p = 0.000$), whereas pressure had a marginal impact. The moisture content (MC%) of different biowaste samples in relation to their particle sizes showing that larger particles maintain higher moisture content with increased variability was presented in Appendix 5(a). The moisture content of the same biowaste samples reveals a pressure-dependent trend in Appendix 5(b) as the applied pressure increases the moisture content decreases.

4.14. Ash content

The ash percentage model output indicates that both sample type ($p = 0.003$) and particle size ($p = 0.001$) were significant variables while pressure ($p = 0.930$) was not significant. The model explains 66.13 % of the variation in ash percentage ($R^2 = 66.13$ %), thus indicating some unexplained variability. Among the sample types, sawdust (SD) has a significant negative effect ($p = 0.000$) on ash percentage by -4.76 units, whereas coffee husk (CH) and paper waste (PW) tend to increase ash content, but not statistically significant. Large particles have a significant positive effect on ash percentage ($p = 0.000$) and medium particles have a significant negative effect ($p = 0.026$) compared to small particle size. Fig. 5 (a) presents the ash content (%) of different biowaste samples as a function of particle size (large, medium, and small). In coffee husk (CH), fruit waste (FrW), and flower waste (FW), larger particles have higher ash content, while in paper waste (PW), sawdust (SD), and vegetable waste (VW), smaller particles have higher ash content. These variations indicate that ash distribution within biowaste is dependent on the material's composition and particle structure.

4.15. Fixed carbon

The results from the model indicated that fixed carbon content was significantly affected by sample type ($p = 0.023$) and particle size ($p = 0.020$), but pressure was not significant ($p = 0.988$). The model explained 57.54 % of the total variation (R^2), which indicates a moderate model fit. Sawdust (SD) had the lowest fixed carbon content among the sample types ($p = 0.027$), and other sample types did not have statistically significant differences. The particle size was also

significant, where large particles had higher fixed carbon content ($p = 0.009$), and medium particles had lower fixed carbon content ($p = 0.018$) than small particles. The larger particles have higher FC% across most of the samples, especially in CH (coffee husk), FW (flower waste), and VW (vegetable waste) which means that coarser fractions retain more carbonaceous material as shown in Fig. 5 (c).

4.16. Calorific value

The General Linear Model (GLM) analysis showed that pressing pressure ($p = 0.300$) is not statistically significant to the calorific value, but sample type ($p = 0.034$) and particle size ($p = 0.001$) are statistically significant. These results agree with the finding of Cavas-Martínez et al. (2021) who established that although the increase in briquetting pressure increases the density and the ignition time, it has no effect on the rate of combustion. The model has a moderate predictive value, accounting for 61.99 % of the total variation ($R^2 = 0.6199$). Among the different forms of biowaste, sawdust (SD) had a significant positive effect on calorific value ($p = 0.002$), medium particle sizes had a positive effect ($p = 0.011$), and large particle sizes had a significant negative effect ($p = 0.001$). Furthermore, the low VIFs indicate that there is no multicollinearity problem, which makes the model more reliable. Fig. 5 (e) shows the calorific values of different biowaste samples and medium and small particle sizes have higher calorific values than large particle sizes. This means that the finer the particle size, the higher the combustion efficiency because of better compaction and burning uniformity. Fig. 5 (f) compares the calorific values at two pressing pressures, and it showed that most biowaste samples had better calorific values at higher pressure.

Overall, the statistical analysis showed that compaction pressure had no significant effect ($p > 0.05$) on calorific value, volatile matter, or fixed carbon. This suggests that thermochemical properties are more strongly influenced by biomass composition and particle size than by compaction force within the tested range.

4.17. Parameter combinations of briquette fuel

The results of parameters showed that sawdust (SD) is the best material to produce high quality briquettes, especially when the particle size is small, and the pressure is 10 MPa (Table 6). This combination had the highest volatile matter (VM) content of 89.06 %, which makes it easier to ignite, and the ash content was low at 1.45 % which means that there will be less residue after combustion. Also, the fixed carbon (FC) content of 19.71 % and calorific value of 19.71 MJ/kg ensures that the burning is sustained, and more energy is produced, hence the highest composite desirability score of 0.6607. This is in conformity with the

Table 6
Optimized properties of briquettes produced from different biowaste materials.

Biowaste	Particle size (mm)	Pressure (MPa)	MC%	Ash%	FC%	Calorific value (MJ/kg)	VM%	Composite desirability
Saw dust (SD)	Small	10	7.97	1.45	19.71	19.71	89.06	0.66
Flower waste (FW)	Small	10	3.62	5.47	13.73	16.55	80.18	0.64
Fruit waste (FRW)	Medium	10	3.30	8.40	13.69	17.15	74.61	0.60
Vegetable waste (VW)	Small	10	7.04	8.66	17.08	16.40	67.23	0.58
Coffee husk (CH)	Small	5	7.16	7.03	10.71	17.62	78.05	0.66

results obtained in Ethiopia where it was established that briquettes made from sawdust were the best in terms of fuel properties as compared to those made from coffee husk, khat waste and dry grass (Kebede et al., 2022). In the same way, Wilson, (2016), established that medium size maize cob particles at 8 MPa made good quality briquettes in Uganda. However, in the present study, small particle sizes with 10 MPa pressure were found to be the most effective, except for fruit waste, which was found to have better quality at medium particle size. The results agree with the finding of Alexandru et al. (2020), who noted that smaller particle sizes help in preventing particle segregation during densification thus improving biofuel quality. The size of the feedstock particles is one of the most important factors that affect the strength and durability of briquettes, with smaller particles producing more compact and stronger briquettes (Narra et al., 2012). Kaliyan & Vance Morey, (2009) also stated that small particle sizes improve compressibility and densification. This has been confirmed by Ibitoye et al. (2022), who found that briquettes produced from carbonized banana stalk and corncob had higher mechanical strength with smaller particle size.

The coffee husk (CH) performed well when processed into small particles under 5 MPa pressure which resulted in 17.62 MJ/kg calorific value and a desirability score of 0.6601 thus making it a suitable biofuel alternative. The Flower waste (FW) and fruit waste (FrW) materials demonstrated average performance through their optimal small and medium particle size combination with 10 MPa pressure which resulted in calorific values of 16.55 MJ/kg and 17.15 MJ/kg. The combustion efficiency suffers from their elevated ash content (5.47 % and 8.40 %) together with their reduced fixed carbon content. The Vegetable waste (VW) required the smallest particle size and 10 MPa pressure for optimization but it contained the highest ash percentage (8.66 %) and produced the lowest calorific value (16.39 MJ/kg) which resulted in the minimum desirability score (0.5827). Research has demonstrated various suitable briquetting pressures which span across three categories of high, medium particle sizes and low-pressure compaction. The finding of (Essien and Oke, 2019), stated that sawdust and rice husk required compacting pressures between 6.86 to 9.81 MPa in low- to medium-pressure machines. Yunusa et al. (2024) predicted that heterogeneous compaction would produce values ranging from 7.10 to 9.75 MPa. The analysis indicates that sawdust together with coffee husk represents the best biomass materials for creating high-quality briquettes but flower and fruit and vegetable waste need blending with better performing materials to improve their fuel characteristics.

4.18. Engineering design and preliminary scale up analysis

To demonstrate the practical applicability and scalability of the proposed briquetting process, detailed engineering calculations are presented below. The calculations focused on four essential aspects, which include die-dimension optimization, residence time estimation, specific energy consumption, and scale-up projections.

a) Die dimension optimization

The cylindrical briquetting die used in this research had dimensions of 50 mm diameter (D) and 110 mm height (H). The die cross-sectional area A which determines the necessary press force can be calculated through $A = \frac{\pi D^2}{4} = \frac{(0.05)^2}{4} = 1.9635 \times 10^{-3} \text{ m}^2$. The pressing

force F needed to achieve 10 MPa (10^7 Pa) compaction pressure is calculated as $F = P \times A = 10^7 \text{ Pa} \times 1.9635 \times 10^{-3} \text{ m}^2 = 19.635 \text{ kN}$. The calculated value matches the operational range of the ZwickRoell Z1200ES press, which demonstrates that the die dimensions are suitable for the system's mechanical capabilities and operational needs.

b) Residence time estimation

The residence time in the briquetting system maintains two essential time periods which consist of 5 min compression followed by 5 min holding for a total of 10 min per briquette. The mold produces six briquettes during each hour of operation. For a setup with multiple molds (n), the production rate reaches $Q = 6n$ briquettes/hour. For instance, using 10 molds simultaneously results in throughput of $Q = 6 \times 10 = 60$ briquettes per hour. This output forms the basis for modelling semi-automated or continuous production system at largest scale.

c) Specific energy consumption

The mechanical energy needed to produce each briquette results from multiplying force by displacement. The energy input calculation requires multiplying the pressing force of 19.635 kN by the stroke (compression depth) of 0.1 m which results in $E = F \times d = 19.635 \times 103 \times 0.1 = 1.9635 \text{ kJ}$. The specific energy consumption for a 60 g or 0.06 kg briquette is calculated as $E_s = 1.9635/0.06 = 32.725 \text{ kJ/kg}$. The energy return on energy invested (EROEI) reaches $19.71 \times 10332.725 = 602.5$, which indicate high energy efficiency under controlled laboratory conditions.

d) Scale-up projection for industrial or institutional use

The conversion of lab-scale results into practical institutional applications required a projection for operational use. The hydraulic press operating with 10 molds for 8 h daily produces 480 briquettes. The daily output reaches 28.8 kg because each briquette weighs 60 g. The small industrial setup with 50 molds produces 300 briquettes per hour, which translates to 2.4 tons per month when operating 8 h daily for 22 days. The output level of this system enables the replacement of 20–25 % of biomass fuel requirements in institutional kitchens through calorific equivalence and current consumption data (965.6 tons/year of charcoal is reported in this study). The total pressing force needed for design purposes grows directly with the number of molds used. The total pressing force needed for 50 molds reaches approximately 1,000 kN. The recommended motor power for each mold station should be between 1.2 and 1.5 kW to account for mechanical losses and operational inefficiencies. The stainless-steel compression chamber provides enough structural strength for producing briquettes at medium production levels. The calculated projections demonstrate that the -briquetting system can be scaled up for use in institutional and industrial waste-to-energy systems.

4.19. Deployment feasibility (Technical and infrastructural perspective)

Ethiopia needs to implement biobased briquettes in practice through methods beyond laboratory evaluation requirements. The paper investigated four essential elements for implementing

the national adoption of briquette technology through system-integration and stove compatibility, as well as user behavior and infrastructure readiness.

a) System integration into the energy sector

Traditional biomass fuels, consisting of firewood and charcoal, serve as the main cooking and heating sources for more than 85 % of the Ethiopian population (Yalew, 2022). The dependence on biomass fuels is most critical in both rural areas and urban informal settlements throughout the country. The production of agricultural residue briquettes serves as a suitable substitute for traditional fuels in urban and peri-urban regions that experience rising charcoal costs and high fuelwood consumption rates, as well as deforestation challenges. The implementation of briquettes matches goals from the Clean Cooking Energy Strategy and the Climate Resilient Green Economy (CRGE) initiative.

b) Technical compatibility with existing stove technologies

The laboratory evaluations demonstrate that the manufactured briquettes match conventional fuels in terms of their calorific value (16.4–20.9 MJ/kg) and ash content (1.6–8.6 %). The physical properties of briquettes match the needs of prevalent Ethiopian cooking devices, including Mirt and rocket stoves for residential and institutional use (Asmamaw and Mulugeta, 2013). Thermal performance and structural stability suggest compatibility with widely used Ethiopian stoves, including the Mirt stove (for injera baking) and institutional rocket stoves. Only minimal stove modifications (e.g., great spacing, air flow control) are expected to ensure optimal combustion. Moreover, the Energy Return on Energy Invested (EROEI) was calculated at 603, indicating very high energy efficiency under controlled compression conditions.

c) User behavior and social acceptance

The adoption of fuels depends on cooking customs, along with user opinions about cleanliness and fuel expenses, and stove usability. Behavioral studies by Beyene and Koch (2013), demonstrate that users will accept the technology if it is designed for ease of use and its price matches charcoal prices. The briquette design mirrors charcoal-products, which will help users accept this fuel type. User acceptance is likely because similar biomass fuels have gained widespread adoption in Ethiopia, but future research requires large-scale trials to confirm this potential.

d) Infrastructure readiness and distribution logistics

The proposed decentralized production and distribution system includes installing briquette production facilities close to the feedstock sources. The close placement of facilities reduces transportation expenses and enables employment opportunities. The existing charcoal distribution network provides a pathway for briquettes to enter the existing energy market. The implementation of targeted investment programs and capacity-building initiatives needs to address infrastructure gaps, which include rural road access, feedstock quality control, and moisture management requirements. The existing clean cookstove dissemination programs in Ethiopia provide a basis to integrate briquette fuels into national energy planning (Kpalo et al., 2020).

4.20. Ecological perspective

4.20.1. Briquetting for waste management

Ethiopia faces severe municipal solid waste (MSW) management issues due to the major cities achieve less than 50 % collection rates, and the national recycling rate stands at 5 % (Endale, 2019; Erasu, et al.

2018). There is high organic content of waste at 70 % of total waste (Gebrekidan et al., 2024), together with the 5 % yearly increase in waste generation. Mohammed & Elias (2017), reported that biomass conversion offers significant opportunities for resource recovery. The situation makes briquetting an appropriate waste-to-energy (WTE) solution. Based on the biowaste audit conducted in hotels and universities of southern Ethiopia, the annual generation of biowaste on dry basis amounts to 26698.93 tons (Seboka et al., 2025), which could create substantial environmental benefits through proper processing of this waste. The waste volume reduction rate measured in this study reached 54.8 % while briquetting can also stop the uncontrolled disposal and decomposition of organic waste that produces greenhouse gases and air pollutants (Munir et al., 2021). In the current study, the energy gained through conversion of one ton of biowastes was 0.294 tonnes of oil equivalent (0.29 toe/ton). This is within range of the finding of Munir et al. (2021), stated that conversion of one ton of waste into briquettes produces between 0.13 and 0.38 tonnes of oil equivalent energy; according to which provides a renewable biomass fuel option for traditional biomass fuels that comprise 85 % of Ethiopia's primary energy mix (Yalew, 2022). The current biomass waste energy production stands at 3 % below its potential which demonstrates a significant underutilization of accessible resources. The implementation of briquetting technology within Ethiopia's waste-to-energy plans would improve waste management facilities while decreasing landfill dependence and climate change impacts and provide more energy access to remote regions.

4.20.2. Deforestation mitigation potential of biowaste briquettes

The extensive use of traditional biomass fuels has resulted in severe environmental damage through deforestation, land degradation, and desertification (Geissler et al., 2013). The Woody Biomass Inventory and Strategic Planning Project (WBISPP) reports that Ethiopia generates 1.2 billion tons of biomass fuel each year from dung and wood, and agricultural residues. The excessive and uncontrolled wood extraction has caused severe forest reduction, which creates energy shortages and elevated wood costs and detrimental effects on climate and agricultural systems. The dryland regions of Ethiopia face severe forest degradation because of fuelwood and charcoal extraction, according to Rahman et al. (2024), who also noted that wood regeneration and biomass increment rates remain low. To evaluate the deforestation mitigation potential of biowaste briquettes, this study utilized data on charcoal and firewood consumption from hotels and universities, based on a previous survey conducted under a similar thematic scope (Seboka et al., 2025). Annual consumption was estimated at 965.62 tons of charcoal and 942,474.7 tons of firewood. According to the model proposed by Chidumayo & Gumbo (2013), using a traditional earth mound kiln efficiency of 0.19 and a woodland biomass stocking rate of 40 tons/ha/year (FAO, 2005), the forest area cleared annually for charcoal production was estimated at 268.3 ha. In addition, based on local biomass stock density, firewood consumption was found to result in 23,561.8 ha of forest cleared each year. In total, the demand for traditional biomass fuels by these institutions contributes to an estimated annual deforestation of 23,830 ha of forests.

In the present study, the analysis of all biowaste types used dry mass measurements through moisture content values from published literature. The corrected dry weights calorific values were received from FAO (2005), Zhang et al. (2007), and Kaliyan and Vance Morey (2009) based on lower heating values (LHV). Real-world combustion losses were reflected through assumed thermal conversion efficiencies. Based on this, the total usable energy potential of the biowaste generated by the surveyed institutions was estimated to 328,045.5GJ/year. By comparison, the total usable energy of the previously quantified annual consumption of charcoal and firewood was approximately 1,613,414.4 GJ/year. While the briquette-based energy yield is lower than the one obtained from traditional biomass. According to FAO (2005), the net calorific value of wood charcoal and firewood was 31 MJ/kg and 18.8 MJ/kg respectively. In this analysis, briquettes can potentially offset up to 20 %

of the current energy demand from forest-based fuels, particularly if the carbonized biowaste is deployed in conjunction with energy-efficient stoves or hybrid systems. Therefore, the partial substitution could reduce pressure on forest resources, offering measurable mitigation of deforestation. However, the deforestation mitigation potential of biowaste-based briquettes in this study is derived solely from laboratory-scale data. Accurate and reliable evaluation will require field trials and comprehensive combustion performance assessments.

The scientific literature reported that briquettes demonstrate equivalent or superior combustion properties when compared to firewood and charcoal. The calorific value of briquettes reaches 7,595 cal/g according to (Sayakoummane & Ussawarujikulchai (2009), which approaches the highest end of wood charcoal values (6,815–7,500 cal/g) (Hassan et al., 2018), and exceeds firewood values (3,500–4,500 cal/g) (Onuegbu et al., 2012). The burning efficiency of briquettes improves because they contain lower moisture (1.88 %) and ash (3.79 %) levels than charcoal. The burning characteristics of briquettes match those of charcoal and agricultural waste sawdust mixtures produce boiling efficiency that surpasses conventional charcoal (Brenda et al., 2017). The current study has not been investigated in the emission of profile of biowaste based briquettes. However, literature reported that emission profile of briquettes shows lower release of greenhouse gases. In line with this, studies indicated that briquettes produce decreased carbon monoxide and carbon dioxide emissions than wood charcoal (Gutema et al., 2024; Kumar et al., 2021). Studies reported that the carbon emissions from floriculture waste briquettes amount to 31.8 % less CO than Acacia charcoal while also stopping the yearly release of more than 16,000 Mg of carbon. The implementation of proper combustion technology helps control the increased particulate emissions that occur with briquettes (Tumutegyeize et al., 2016). The combination of environmental advantages and health benefits together with forest conservation makes briquettes a vital component in Ethiopia's energy transformation

5. Limitations and future research

The current research provided important findings about biowaste briquetting parameter optimization and demonstrates improvements in fuel quality and mechanical durability, yet several important limitations need to be recognized. The Binder ratio results were based on preliminary trials (3:1, 4:1, 5:1) assessing density, calorific value, ash content, and durability. A full sensitivity analysis, including 2:1 ratios and energy balance calculations, was beyond the current study's scope and is recommended for future work. In the current study, the interpretations were restricted to macroscopic observations and measured bulk properties, avoiding extrapolation to microscale features, (e.g. SEM). Future work should incorporate microstructural characterization to validate the mechanistic explanations and strengthen the link between compaction behavior, pore morphology, and combustion performance. The laboratory-based experiments do not fully reflect the complete range of operational variations which occur during field applications. The study did not perform emissions analysis through TGA and DSC and gas pollutant measurements which made the evaluation of combustion-related environmental impacts preliminary. This work was also not conducted crucial assessments of environmental footprint and economic viability and scalability through life cycle assessment (LCA) and techno-economic analysis (TEA). The study included preliminary scale-up estimates, yet it did not perform pilot-scale testing or long-duration performance trials to evaluate throughput capacity and energy efficiency and equipment reliability during continuous operation. Future research needs to fill these knowledge gaps through complete emissions profiling and LCA and TEA modeling and field-based trials that evaluate briquette combustion behavior and stove compatibility and user acceptability in actual usage conditions.

6. Conclusion

This study examined how biomass type, together with particle size- and compaction pressure, affects the thermochemical and physical properties of biowaste briquettes by using pulp paper as a binder. The research showed that briquette quality depends significantly on both particle size and biomass type, but compaction pressure had a limited impact on the results. The combination of small sawdust particles with 10 MPa pressure produced briquettes that achieved optimal fuel characteristics through their high calorific value, low ash content, and strong structural integrity. The addition of the binder enhanced both combustion efficiency and dimensional stability, and durability tests showed that the briquettes maintained strong mechanical properties, confirmed through drop and abrasion tests. The research indicated that biowaste briquetting holds promise as a sustainable waste management and energy solution, but multiple scientific and practical challenges need further investigation. The study did not evaluate combustion emissions, conduct a life cycle or techno-economic analysis, or confirm performance under operational field conditions. Therefore, anticipated environmental and deforestation mitigation benefits, while promising, require further research and pilot-scale validation to confirm feasibility and impact at larger scales.

CRediT authorship contribution statement

Ararsa D. Seboka: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **John Morken:** Writing – review & editing, Validation, Supervision. **Muyiwa S. Adaramola:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Gebresilassie A. Ewunie:** Writing – review & editing, Supervision, Methodology. **Lu Feng:** Writing – review & editing, Supervision, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2025.133277>.

Data availability

Data will be made available on request.

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